Winning Back the Base

A Data-Driven Customer Retention Strategy for Company A

Predicting and preventing telecom churn with machine learning — turning 100,000 customer records into a quantified, ROI-positive retention program.

0.693
XGBoost ROC-AUC

$22.6M
net benefit / yr (per 1M subs)

2.57x
return on retention spend

IT Consulting — Proof of Concept · Prepared for Company A Executive Team

Author: [Omnicampus Account Name] · June 2026

Agenda

A saturated market where growth means stealing share

$344B
U.S. wireless market size (2025)

~1.0%
Postpaid monthly churn, best-in-class

5–25x
Cost to acquire vs. retain a customer

25–95%
Profit lift from +5% retention (Bain)

The U.S. wireless market is large but saturated: carriers compete for a fixed pool of subscribers, so almost all net growth comes from winning competitors' customers rather than new demand. In that environment, every customer who leaves must be re-acquired at 5–25x the cost of keeping them.

Industry research is unambiguous: small improvements in retention compound into large profit gains. A 5% increase in retention can lift profits 25–95% (Reichheld / Bain), and customer-level retention models have been shown to materially increase carrier profitability. Retention — not acquisition — is the highest-ROI growth lever available to Company A today.

Sources: IBISWorld U.S. Wireless Telecom 2025; Verizon/T-Mobile investor reports 2024-25; Reichheld & Bain & Co.; Harvard Business Review. Full list on slide 15.

The client and the data we were given

Company A is an anonymized telecommunications provider. It holds a large customer dataset but limited in-house analytics capacity. For this Proof of Concept we received two linked tables covering ~100,000 customers:

Client.csv — who the customer is

Demographics & account profile: handset price, device age, credit class, tenure, plan attributes, household data. 50 columns.

~100,000 rows

Record.csv — how they behave

Usage history: monthly revenue, minutes of use, recurring charges, overage, call quality — and the churn label. 51 columns.

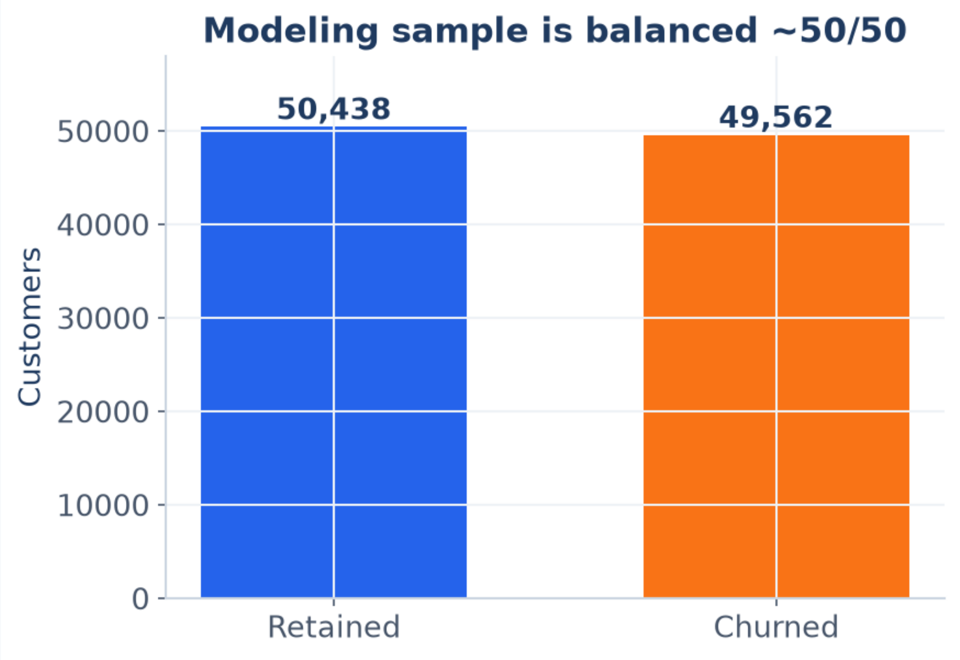
+ the churn flag

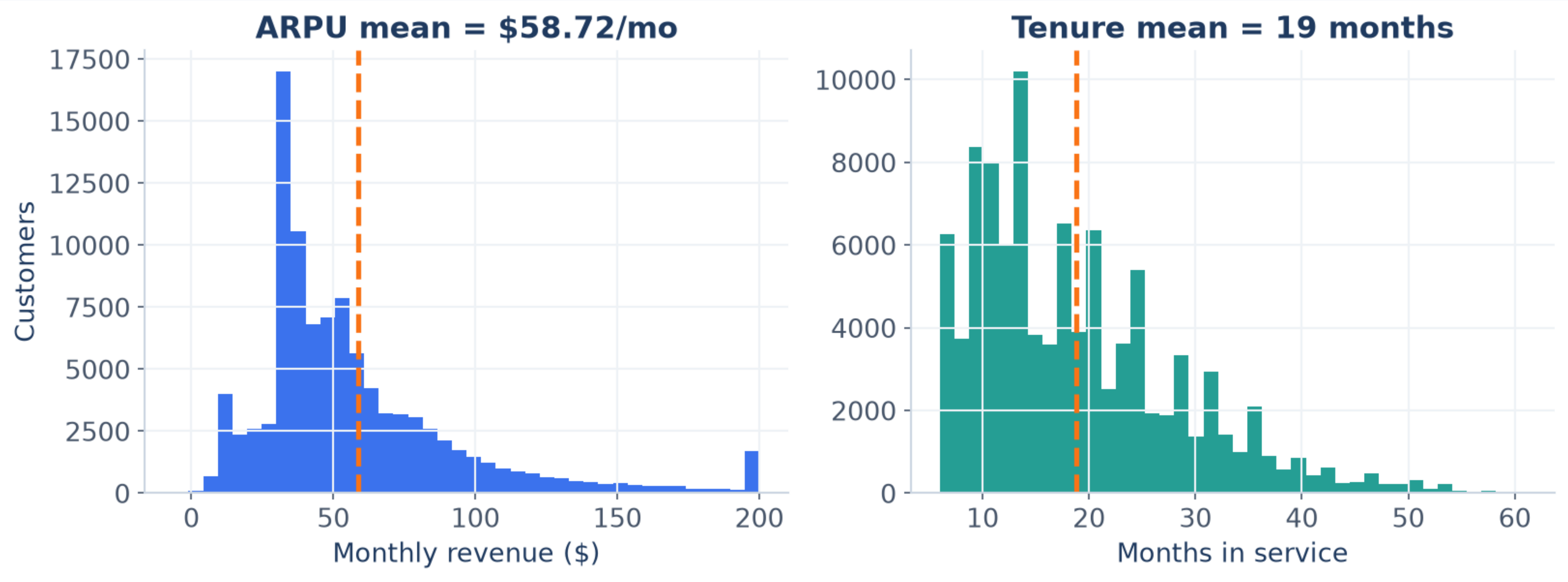
Tables join on Customer_ID → one 100,000 × ~100 modeling table.

Real-world constraint: some columns are undocumented and many profile fields are sparse. Handling that ambiguity is part of the engagement — we lean on the well-populated, business-meaningful features.

Source: Company A Dataset Overview (provided). Analysis: authors' merge of Client.csv + Record.csv.

Each customer is worth ~$59/month — and they don't stay long

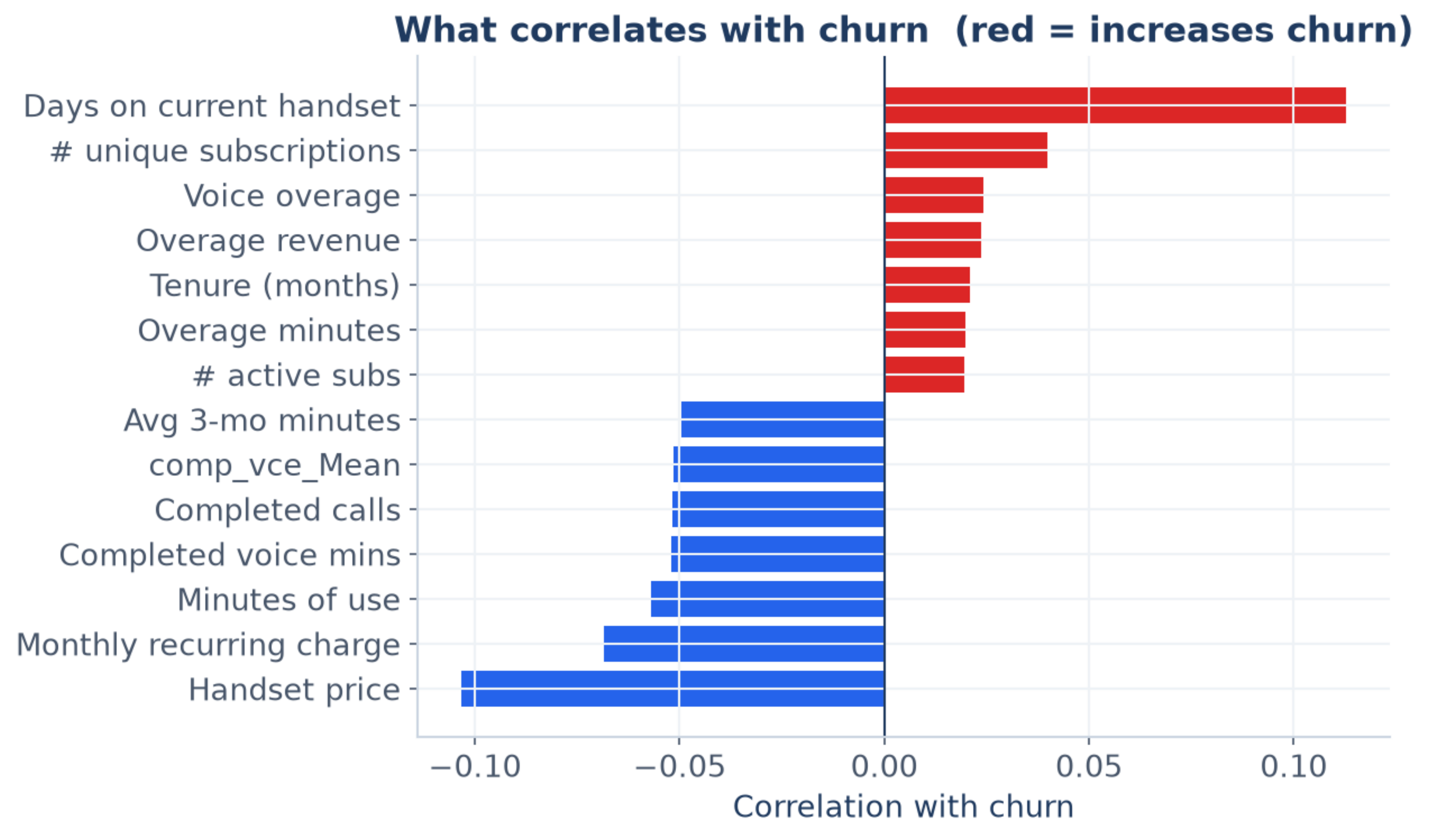




Takeaway: the modeling sample is deliberately balanced 50/50 (not the real churn rate). Average revenue per user is $58.72/mo and median tenure is just 16 months — so losing a customer early destroys significant lifetime value. This is the value at stake.

Analysis: authors' EDA on merged dataset (n=100,000). ARPU = rev_Mean.

Aging handsets drive churn; newer / pricier devices retain



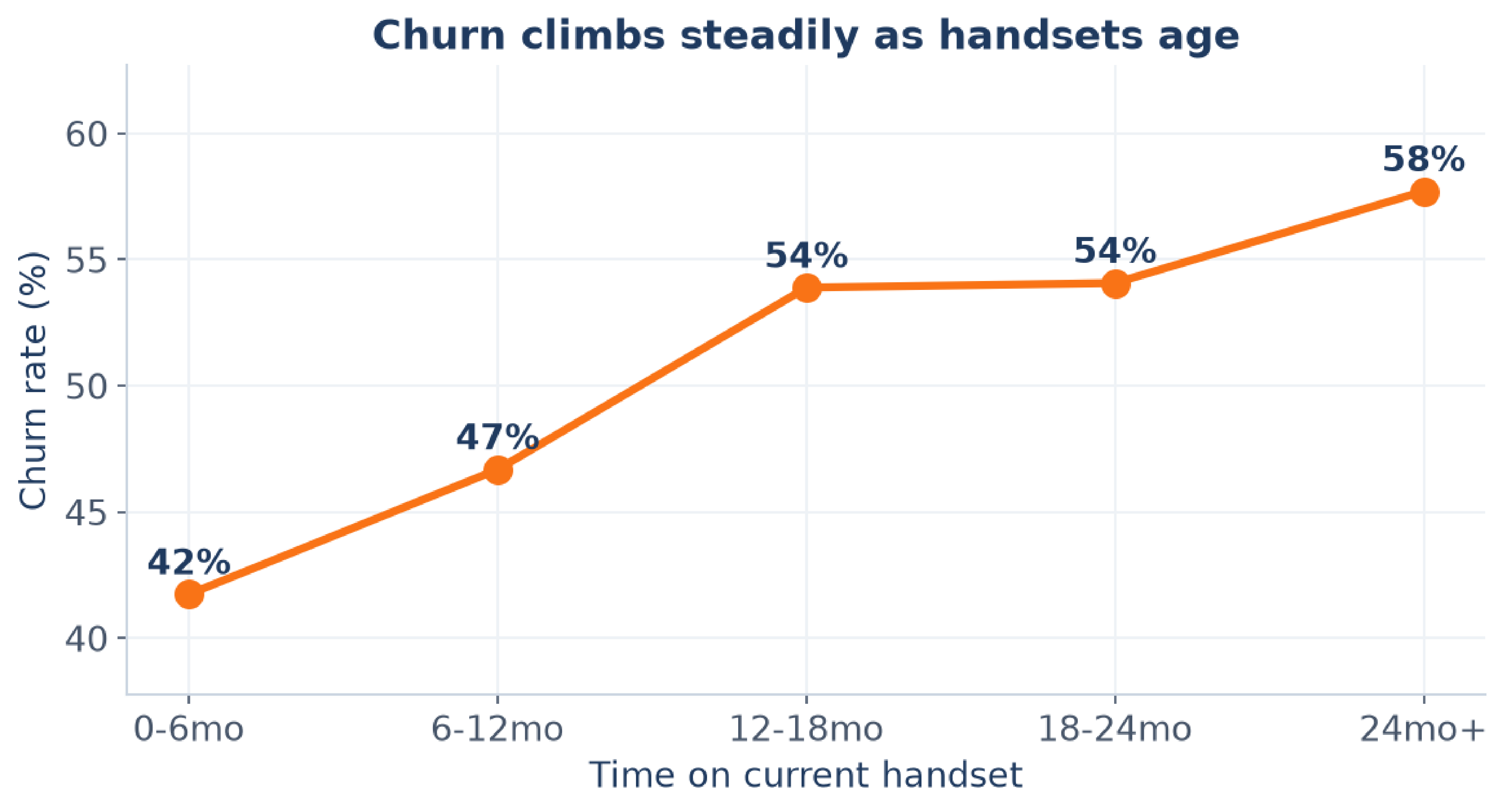
What the correlations tell us

- Days on current handset is the strongest churn-increasing signal.
- Handset price & monthly charge are the strongest churn-reducing signals.
- Higher engagement (minutes of use) tracks with staying.
- Customers with more/older equipment are flight risks.

Hypothesis: customers stuck on old devices feel “stale” and are courted by rivals' upgrade offers. Device lifecycle is a lever Company A directly controls.

Analysis: authors' point-biserial correlations of numeric features with churn.

The headline insight: churn rises monotonically with handset age



lowest

Churn, handsets < 6 months old

highest

Churn, handsets 24+ months old

Why it matters

Equipment age is observable, predictable, and actionable. We can see a customer aging into the danger zone months ahead — and intervene with an upgrade offer before a competitor does.

Analysis: authors' EDA — churn rate by eqpdays (days on current handset) bins.

PROBLEM DEFINITION

From insight to a precise machine-learning question

Business objective: reduce voluntary churn by identifying at-risk, high-value customers early enough to act.

ML TASK

Supervised binary classification

Predict P(churn) for every customer.

TARGET VARIABLE

churn (1 = left, 0 = stayed)

Chosen from EDA + market: retention is the #1 profit lever.

PRIMARY METRIC

ROC-AUC

Ranks customers by risk, threshold-free, robust to the balanced sample.

OPERATING METRIC

Lift / gains

Who to call first — the metric the campaign actually runs on.

Rationale grounded in EDA (slides 5-7) and market analysis (slide 3).

How we built it — explained without the jargon

1 Clean & join

Merge the two tables on Customer_ID; remove IDs to avoid cheating.

2 Engineer features

Impute missing values; scale numbers; encode categories — 250+ model inputs.

3 Train 3 models

A simple baseline and two advanced tree models, on 75% of data.

4 Test fairly

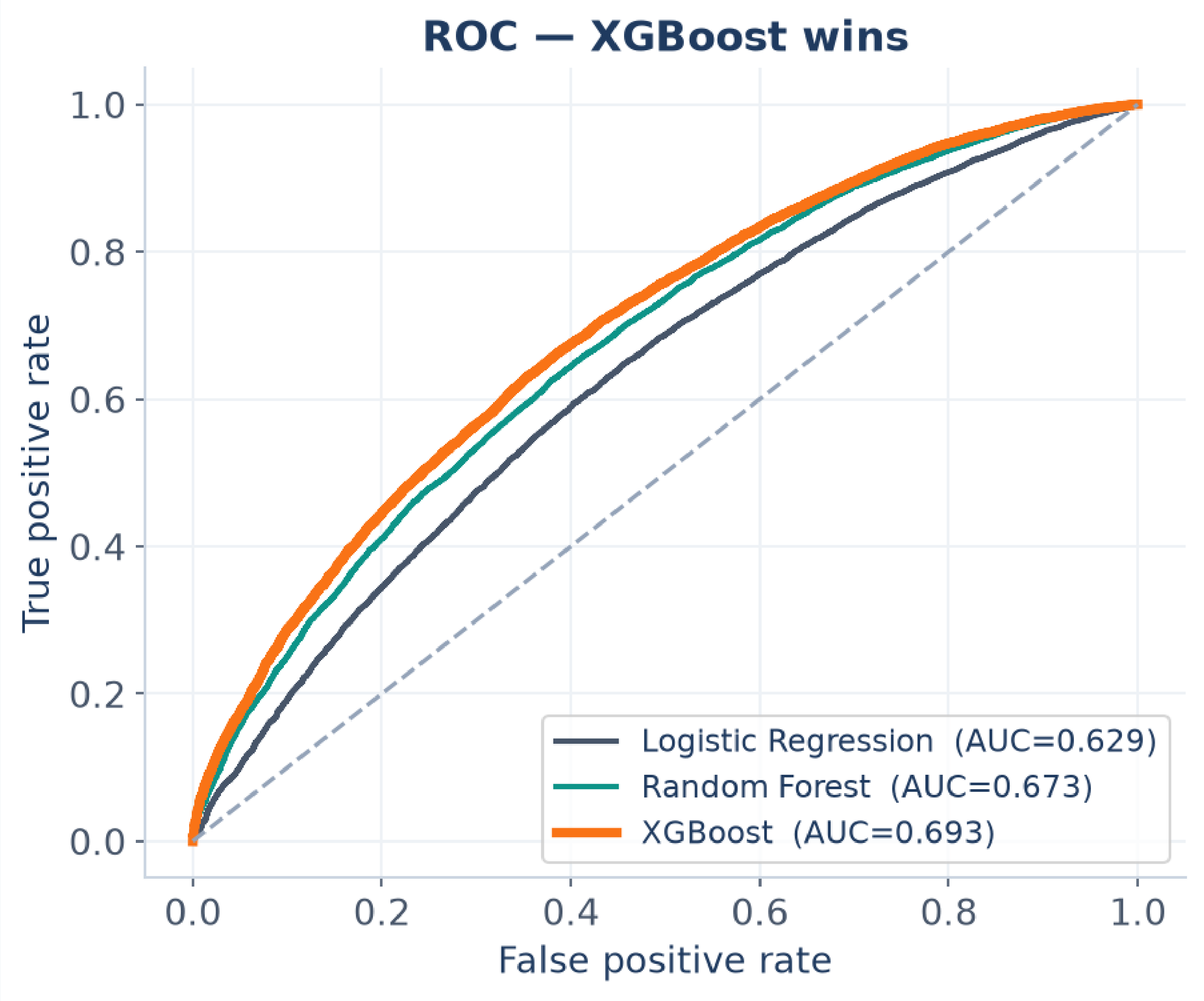
Score the held-out 25% the model never saw — honest performance.

Why these models?

Logistic Regression gives a transparent, explainable baseline. Random Forest and XGBoost capture non-linear interactions (e.g. “old handset AND low usage”) that drive churn. XGBoost is the industry standard for tabular data — so we let the data pick the winner rather than assuming one.

Method: scikit-learn pipelines + XGBoost. Reproducible in the accompanying notebook. Ref: Chen & Guestrin 2016 (XGBoost).

Results: XGBoost is the best model — AUC 0.693



Model performance (held-out test set)

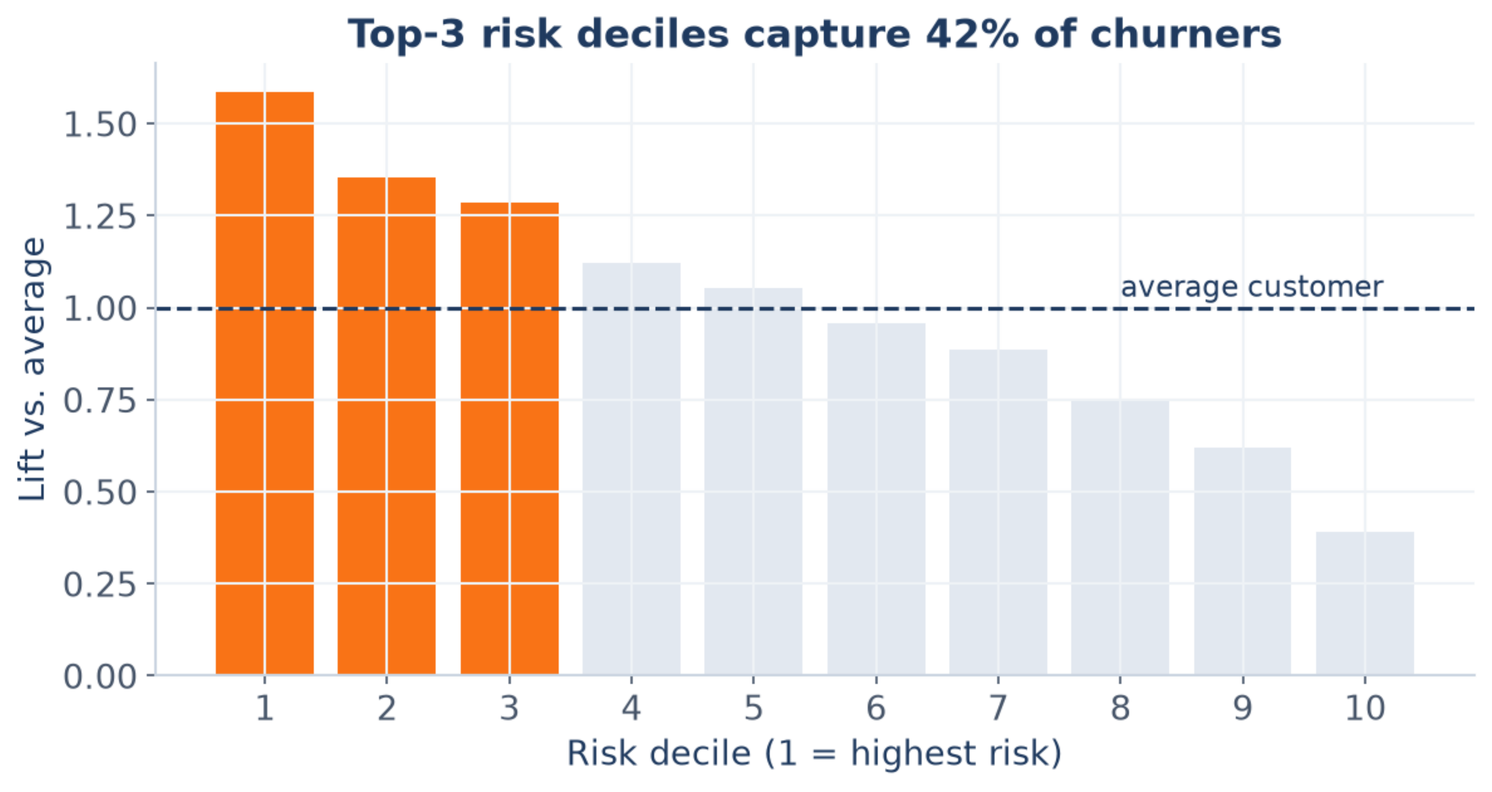
Model	AUC	Precision	Recall	F1
Logistic Regression	0.629	0.592	0.590	0.591
Random Forest	0.673	0.612	0.650	0.631
XGBoost	**0.693**	**0.631**	**0.645**	**0.638**

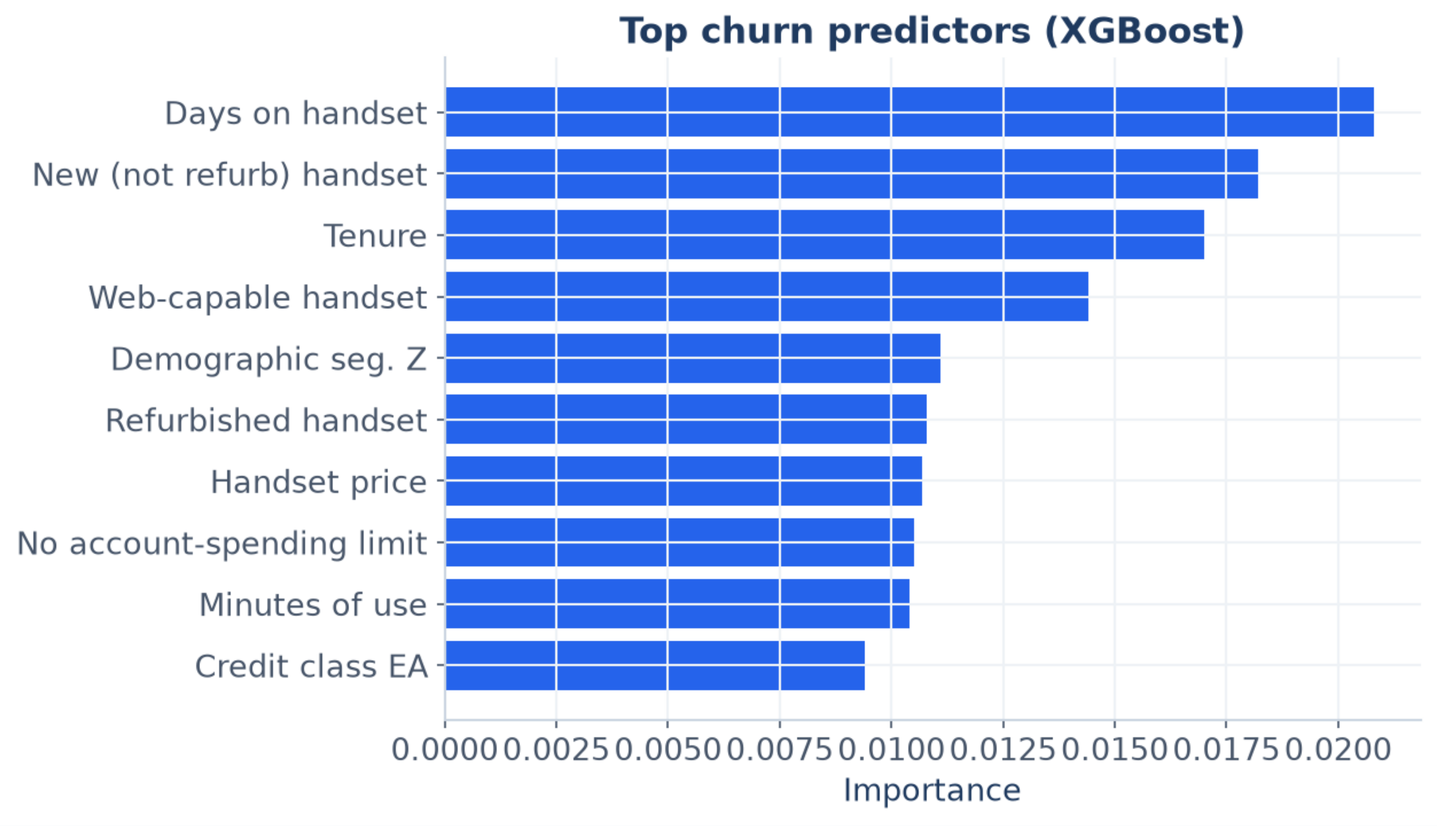
Winner: XGBoost — chosen for deployment.

How to read AUC: 0.5 = random guessing, 1.0 = perfect. At 0.693 the model reliably ranks a churner above a non-churner ~69% of the time — strong enough to target retention spend profitably.

Metric: ROC-AUC (primary) + precision/recall/F1. Scores from authors' notebook, 25% hold-out, seed=42.

The model tells us WHO to call — and WHY they're leaving





Two model outputs the business can act on directly:

1) LIFT — Ranking customers by risk and contacting only the top 30% reaches 42% of all churners. That is up to ~1.6x more efficient than blanket outreach, so every retention dollar works harder.

2) DRIVERS — The top predictors (handset age, new-vs-refurbished device, tenure, device web-capability) are exactly the levers Company A controls. The model doesn't just flag risk — it points to the fix: proactive device upgrades.

Analysis: authors' lift/gains on test set; XGBoost feature importances.

BUSINESS PROPOSAL

The proposal: a model-driven proactive retention engine

Score every customer monthly, then act on the highest-risk, highest-value segment before they leave.

Predict

Run the XGBoost model monthly across the full base to produce a churn-risk score for every customer.

Prioritize

Rank by risk × value. Focus retention budget on the top deciles — where 42% of churn concentrates.

Intervene

Trigger the right offer: proactive handset upgrades for aging-device customers, loyalty perks for the rest.

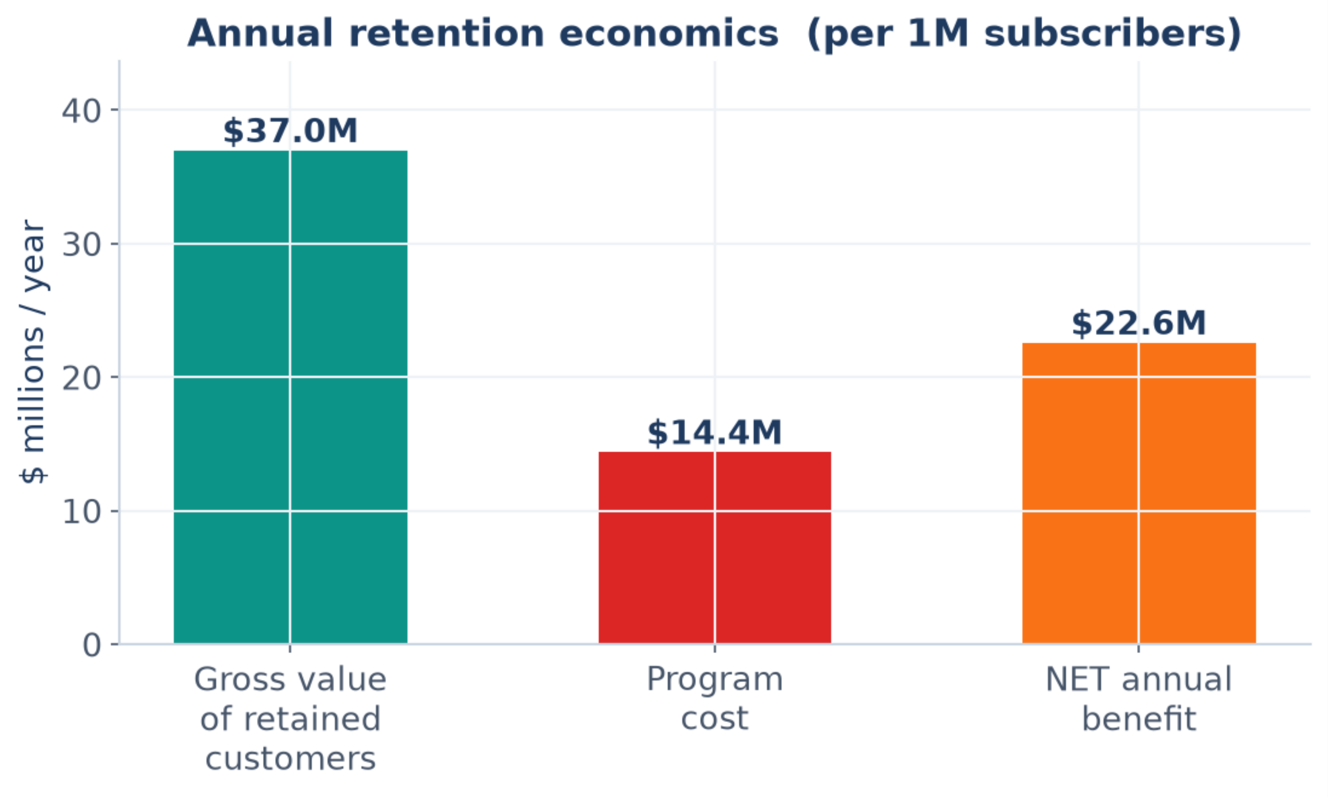
Measure & learn

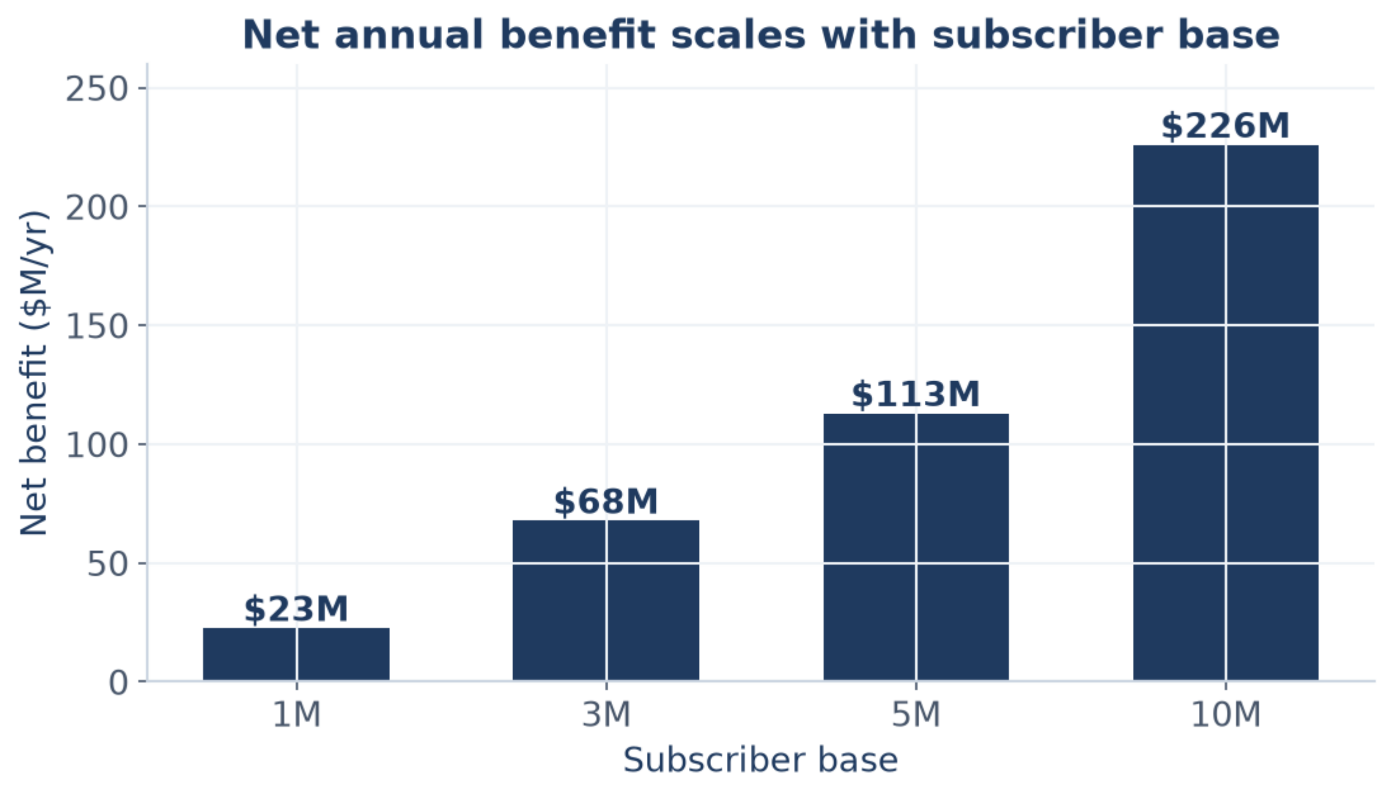
Run as a treatment-vs-holdout A/B test to prove incremental saves, then scale what works.

The core fix the data points to:

Equipment age is the #1 churn driver AND a lever Company A owns. A targeted proactive-upgrade program attacks the root cause — not the symptom.

Quantified impact: $22.6M net benefit per million subscribers





Assumptions (conservative, stated explicitly)

- Operating churn: 2.0% / month (challenged carrier)
- ARPU: $58.72 / mo (measured) · Gross margin: 50%
- Contribution CLV: ~$1,468 per customer
- Target top 30% risk · captures 42% of churners
- Save rate: 25% of contacted would-be churners
- Program cost: $4 per contacted customer

Result: churn 2.0% → 1.79%/mo · ROI 2.57x · payback < 1 yr

Scales linearly: a 5M-sub carrier ≈ $113M net benefit / yr

Authors' model; CLV = margin ÷ monthly churn. Retention-economics benchmarks: Bain/Reichheld; HBR. Full derivation in notebook §6.

NEXT STEPS

Roadmap, risks, and how we de-risk the rollout

A staged rollout that proves value before full investment.

Phase 1 — Validate

Wks 1-6

Lock the data feed, deploy scoring, design the A/B retention test on a single high-risk segment.

Phase 2 — Pilot

Wks 7-16

Run treatment-vs-holdout upgrade offers; measure true incremental save rate and ROI.

Phase 3 — Scale

Q3+

Automate monthly scoring across the base; integrate offers into CRM; add real-time triggers.

Risks & mitigations

- **Model captures correlation, not causation** → Validate lift with a randomized holdout before scaling spend.
- **Sparse / undocumented profile fields** → Rely on well-populated behavioral features; refresh data dictionary with client.
- **Offer cannibalization (discounts to loyal customers)** → Target by risk × value; suppress low-risk customers via the score.
- **Model drift over time** → Re-train quarterly; monitor AUC and realized save rate.

References

1. IBISWorld (2025). Wireless Telecommunications Carriers in the US — Market Size & Industry Statistics.
2. Verizon Communications (2025). Q4 2024 / FY2024 Earnings Report — postpaid churn & ARPU disclosures. verizon.com/about/investors
3. T-Mobile US (2025). Q4 2024 Investor Factbook — postpaid phone churn. t-mobile.com/news
4. Reichheld, F. & Sasser, W.E. (1990). “Zero Defections: Quality Comes to Services.” Harvard Business Review.
5. Bain & Company / Reichheld, F. — Prescription for Cutting Costs (customer retention & profitability).
6. Gallo, A. (2014). “The Value of Keeping the Right Customers.” Harvard Business Review.
7. Neslin, S. et al. (2006). “Defection Detection: Measuring and Understanding Predictive Accuracy of Customer Churn Models.” Journal of Marketing Research.
8. Chen, T. & Guestrin, C. (2016). “XGBoost: A Scalable Tree Boosting System.” KDD '16.
9. Pedregosa, F. et al. (2011). “Scikit-learn: Machine Learning in Python.” JMLR 12.
10. Company A Dataset Overview & Final Assignment README (provided course materials), GCI World 2026.
11. Dataset: Cell2Cell-style telecom churn data (Client.csv + Record.csv), provided by the course.

Generative AI (ChatGPT/v0) was used to assist drafting and diagramming; chat URL provided in the Omnicampus references field. All analysis, code, and figures are the authors' own.